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6. In one example of the process of nuclear fission of uranium-235, it is suggested that isotopes of caesium and rubidium and a number of neutrons ($^{1}_{0}n$) are produced.

The table gives information about the particles in the nuclei of the isotopes named above.

Element	Symbol	Number of protons in the nucleus	Number of neutrons in the nucleus
uranium	U	92	143
caesium	Cs	55	85
rubidium	Rb	37	55

(a) Write a balanced nuclear equation for the fission decay of uranium into caesium and rubidium. [4]

$$^{\cdots\cdots}_{\cdots\cdots}U \;+\; ^{1}_{0}n \longrightarrow \;\cdots\cdots\cdots\; + \;\cdots\cdots\cdots\; + \;\cdots\cdots\; ^{1}_{0}n$$

(b) (i) Explain the purpose of the moderator in a fission reactor. [2]

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(ii) Explain how the rate of reaction can be increased in a nuclear reactor. [3]

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(c) The caesium isotope produced in this fission reaction has a half-life of 64 s.
The rubidium isotope has a half-life of 4 s.
At time, $t = 0$ the mass of caesium present is 131 072 g.
The same mass of rubidium is also present.

(i) Calculate the mass of caesium present after 64 s. [1]

mass of caesium = ... g

(ii) I. Calculate how many half-lives of rubidium occur in 64 s. [1]

number of half-lives = ...

II. After 32 seconds, 512 g of rubidium remains.
Calculate the mass of rubidium remaining after 64 s. [3]

mass = ... g

(iii) Initially the ratio of mass of caesium to mass of rubidium was 1:1.
Explain how this ratio changes with time. [2]

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